UGANDA CHRISTIAN UNIVERSITY

A Centre of Excellence in the Heart of Africa

Faculty of Engineering, Design and Technology
Department of Computing and Technology

# Agricultural Services Database System

Database Programming (CSC2209)

Project-Based Examination
Easter Semester 2026 — BSCS Year Two Sem Two

**Group 11**

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# 1 Introduction

The Agricultural Services Database System provides the Ministry of Agriculture with a single centralized source of information to help farmers manage their coffee farms instead of using a dispersed manual record keeping system. The system provides an efficient means for farmers to store, retrieve and analyze agricultural data. As such, it will enable the Ministry to better manage and deliver services to farmers by ensuring that all agricultural operations are tracked in an organized manner and that data is consistent and reliable.

# 2 Objectives

The primary goal of this project is to design and implement a managed database system that provides a usable and secured source of information to aid operational management of agriculture, in this case coffee production, basing off the following objectives:

- To design a database for managing farmers and farms
- To track coffee production and sales
- To manage agricultural support distribution
- To facilitate interaction between farmers and extension workers
- To ensure data integrity and security

# 3 Functional Requirements

This system will provide for the necessary functions of any agricultural activity to occur within the agricultural system for example it allows to:

- Register persons (farmers and extension workers)
- Assign farmers to cooperatives
- Record farms and ownership
- Track coffee production and sales
- Manage support distribution
- Record farm visits and pests
- Manage training sessions and attendance
- Enforce user roles and permissions

# 4 Assumptions

These following assumptions simplify the design of our database while maintaining alignment with real-world agricultural practices.

- Each farmer belongs to exactly one cooperative
- Each farmer owns at least one farm
- Coffee types are either Arabica or Robusta
- Training sessions are categorized as field, workshop, or online

# 5 Database Design (EERD)

To create an Enhanced Entity Relationship Diagram (EERD) for the agricultural domain, we established a way to model the various entities (objects), relationships and constraints associated with the agricultural sector. Our design provides a realistic representation of the various relationships that exist in the real world between farmers, farms, extension workers and agricultural services while providing for logical consistency.

The primary entities in the system are Person, Farmer, Extension Worker, Cooperative, Farm, Coffee Type, Production, Sales, Support and Distribution, Training and an associated Location (Region, District, Sub-County). Each entity in the database will represent a separate piece of information about the agricultural sector, and will also contain attributes that relate to that entity

## 5.1 Specialization

A Generalization of the Person entity has been created as a super class of the Farmer and Extension Worker entities, allowing for an overlap of specialization, with one Farmer/Extension Worker being able to perform multiple roles. The disjoint specialization for the Coffee Type, Support Type and Training entities ensures that an instance can belong to only one entity within that classification.

**Overlapping:**

- Person → Farmer, Extension Worker

**Disjoint:**

- Coffee Type → Arabica, Robusta
- Support Type → Input, Financial, Training
- Training Session → Field, Workshop, Online

## 5.2 Key Relationships

The entities in the EERD were defined based on real-world constraints between the two types of entities. These relationships were utilized in developing the EERD and they are as follows:

- Farmer owns Farm (1:N)
- Extension Worker registers Farm (1:N)
- Farm produces Coffee (1:N)
- Farm grows Coffee Type (M:N)
- Farmer attends Training (M:N)

# 6 Database Implementation

The implementation of the database was done with MySQL 8.0, and tables were created for every entity defined in the design stage. Primary keys were used as the keys that identify each record. Foreign keys established the relationships between the different tables and also helped maintain data integrity through referential integrity.

To represent many-to-many relationships, junction tables were created to ensure that the database is normalized, and redundancy is minimized.

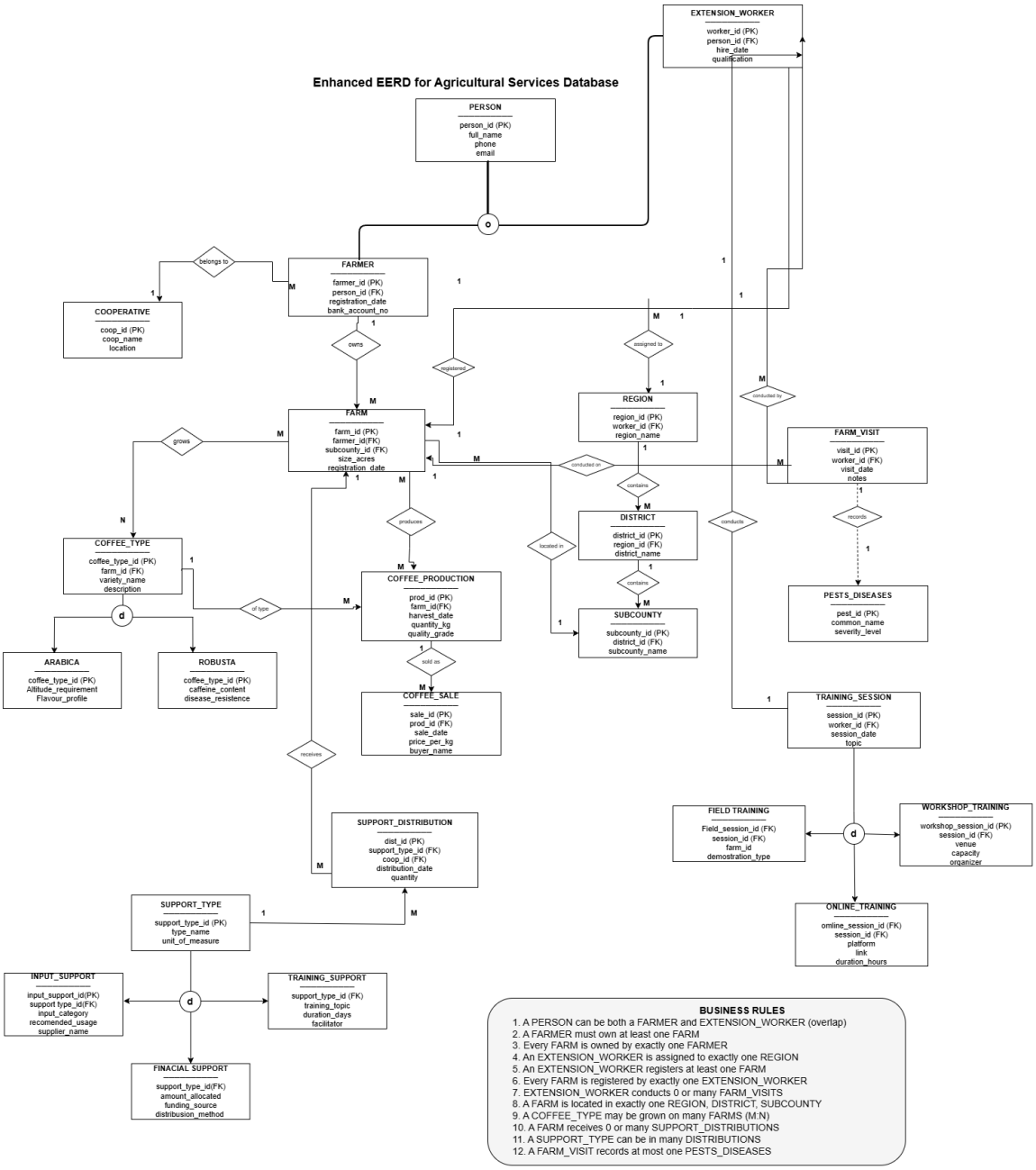


Figure 1: Enhanced Entity Relationship Diagram of the System

## 6.1 Structure

The system consists of multiple tables with primary keys and foreign keys enforcing relationships.

## 6.2 Constraints

Through the use of constraints such as NOT NULL, UNIQUE, and CHECK conditions, data validation was enforced. These constraints help to ensure that the only valid data is stored in the database and that no inconsistencies exist.

- Primary Keys ensure entity integrity
- Foreign Keys enforce relationships
- CHECK constraints validate data values

Example:

```
CHECK (quantity > 0)
```

## 6.3 Triggers

Triggers were implemented to enforce complex business rules which basic constraints cannot deal with. For example, a trigger was used to ensure the coffee sold did not exceed the total quantity produced. In addition, triggers were used to ensure that the specialty of disjoint specialization is achieved, and to prevent invalid deletions, thus maintaining a healthy database.

- Preventing sales exceeding production
- Enforcing disjoint specialization
- Preventing deletion of critical data

# 7 Security and Automation

## 7.1 User Roles and Role Based Access Control

Role-based access control was used in MySQL to enforce security. Various roles have been defined, and each role has a specific set of privileges. This ensures that users may only perform actions related to their responsibilities and they include:

- Administrator

- Extension Worker
- Data Entry Clerk
- Viewer

### 7.2 Stored Procedures

Stored procedures were made for automating common tasks such as registering farmers, adding farms, keeping track of production, providing support, and logging farm visits. Stored procedures encapsulate logic, thereby improving efficiency and reducing errors and ours included:

- Register Farmer
- Add Farm
- Record Production
- Distribute Support
- Record Farm Visit

### 7.3 Backup and Recovery

The system supports backup and recovery through database export and import mechanisms. Backup files can be created and used to restore the database in case of failure, ensuring data reliability.

## 8 Testing and Results

The complete project, including the database scripts and supporting files, is hosted on GitHub and can be accessed via the following link:

$$\texttt{https://github.com/obyelo/database}_{e}xam_{g}roup11$$

### 8.1 Testing

Sample data was used to test the system's functionality. Multiple test cases were run to validate the constraints, triggers, and relationships to ensure the system performs correctly under various scenarios

### 8.2 Sample Queries

```
SELECT * FROM farmer_summary;
SELECT * FROM production_summary;
SELECT * FROM support_distribution_report;
```

### 8.3 Results

The testing showed that the system enforces all the constraints and business rules defined by the System. The triggers prevented illegal operations, and the role-based access successfully restricted unauthorized access. The system was shown to be reliable and consistent in its handling of agricultural data.

- All constraints function correctly
- Triggers enforce business rules
- Roles restrict unauthorized access

## 9 Challenges

Complex business rules using database level mechanisms like triggers and constraints presented one of the primary challenges while developing this system. Particularly difficult was ensuring that both disjoint and overlapping specializations were correctly enforced via a well-designed and tested process when working with tables containing subtypes and ENUM based classifications as well. Another challenge was keeping referential integrity in multiple related tables, especially during data insertions, where the order of operations had to follow strict dependency relationships.These various challenges were overcome by means of iterative testing, schema restructuring when needed, and using triggers and stored procedures to consistently enforce rules throughout the system.

## 10 Conclusion

The database system that was developed offers a complete and scalable solution for managing agricultural services, integrating the design, implementation, and security functions together to meet the operational requirements of the Ministry of Agriculture. The system provides data integrity, increases efficiency, and gives the Ministry a dependable method to manage agricultural data.